Lesson Plan · Tuesday, November 10

Algebra I · Mr. Pandey · Module 3, Topic 1, Lesson 2: Literal Equations (Day 2 of 2) · essentials only (✪)

Objectives (students will…)

- Define literal equation.
- Solve formulas for a specified variable and justify the steps.
- Use a rewritten formula to find an unknown measure.

TEKS

- **A.12E** solve formulas and other literal equations for a specified variable
- A.3A, A.2B
- Process: A.1A

Key terms

- literal equation

Materials

- Slides: m3t1-lesson2-literal-equations (.html/.pptx) · publisher slides with answers: BLMathematics_AlgI_M3_T1_L2_LessonSlides_ENG_with_answers.pptx
- Bell work: BELL WORK/T1-Bell-Work-Tue-11-10.pdf + key (online: m3t1-lesson2-bellwork-day2)
- Exit ticket: EXIT TICKET/T1-Exit-Ticket-Tue-11-10.pdf + key (written like the End of Topic 1 test)
- Video: videos/M3T1-Lesson2-Literal-Equations.mp4 · IEP and 504 versions of all of these: IEP/, Section 504/, and the -IEP / -504 PDFs
- Student Edition pp. 362–367 (Activities 2.2, 2.3, Talk the Talk)
- Calculators (3.14 for π)
- Whiteboards

Bell work (on the board)

- Which equation is 4x + 2y = 10 written in slope-intercept form? B. y = −2x + 5
- What value of x makes 3x − 7 = 11 true? H. 6
- P = 2ℓ + 2w. What is ℓ when P = 30 and w = 5? D. 10
- C = 5/9(F − 32). What is C when F = 50? F. 10

Agenda (60 minutes)

Time	Activity	What to do / say
0–5	Bell work	Q3–4 lead into today: we keep solving the same formula for different letters.
5–25	✪ Activity 2.2: Rewriting Literal Equations · SE pp. 362–364	Define *literal equation*. Q1 solve C = 5/9(F − 32) for F: F = 9/5 C + 32. Q2 77°F vs 30°C: Mateo is warmer: 30°C = 86°F. Q3 error analysis: Ethan didn't distribute 5/9 to the 32; Minh multiplied by 9 wrong (45F, 1440). Worked Example (clear fractions), then Q4: 9C − 5F = −160.
25–45	✪ Activity 2.3: Solving Literal Equations for Specific Variables · SE pp. 365–366	Q1 V = Bh → h = V/B; V = 450, B = 75 → 6 in. Q2 cone V = ⅓πr²h → h = 3V/(πr²). Q3 trapezoid A = ½h(b_1 + b_2) → h = 2A/(b_1 + b_2); A = 24, bases 9 and 7 → 3 cm. Q4 Z = A/B + C/D for A: A = BZ − BC/D (part b, solve for D, is the challenge). **Test move** (publisher slide 52, like test Q8): V = ℓwh for w w = V/(ℓh); A = ½bh for h h = 2A/b.
45–57	✪ Talk the Talk: Rearrange It · SE p. 367 (exit ticket)	1. The formula for the area of a rectangle is A = ℓw. Solve the equation for w. w = A/ℓ 2. The formula for the volume of a cylinder is V = πr²h. Solve the equation for h. h = V/(πr²) 3. Which equation is d = rt solved for r? r = d/t 4. S = 2πrh, S = 37.68, r = 1.5. Find h. h = 4 in. IEP/504: read aloud on request, properties/notes card, extra time.
57–60	Wrap up	Essential Question (SE p. 357). Homework: Lesson 2 Assignment Practice 2–3 and Prepare (SE p. 370). Prepare previews inequalities.

Check for understanding

- Whiteboards: "Solve A = ℓw for w." w = A/ℓ
- Thumbs: "To get F alone, subtract 32 first." Down: multiply by 9/5 first
- Activity 2.3 Q1–4 are the TE's check points.

After class

- Missed Talk the Talk 1 or 3 → Skills Practice Set D (literal equations).
- Missed 2 or 4 only → calculator/substitution check, small group.
- Prepare answers: x = 5, x > 7, x ≤ −3, x < −4.

All of Day 2 is starred. Prepare Q4 (−x > 4 → x < −4) sets up flipping the sign in Lesson 3.